Auteur: Viki Lauvrys
Studierichting: Informatica
Oefeningenreeks 1E nr. 10

$$A(z) = z^3 - 3iz^2 + (1+i)z - 1 + 2i$$

$$B(z) = (z-i)(z+2i)$$

$$A(i) = i^3 - 3i(i^2) + (1+i)i - 1 + 2i = -2 + 5i$$

$$A(-2i) = (-2i)^3 - 3i(-2i)^2 + (1+i)(-2i) - 1 + 2i$$

$$\Rightarrow \begin{cases} ai + b & = & -2 + 5i \\ -2ai + b & = & 1 + 20i \end{cases}$$

$$\Rightarrow \begin{cases} \dfrac{i - 20 - bi}{2} i + b & = & -2 + 5i \\ a & = & \dfrac{i - 20 - bi}{2} \end{cases}$$

$$\Rightarrow \begin{cases} b & = & -2 + 5i - \dfrac{-1 - 20i + b}{2} \\ a & = & \dfrac{i - 20 - bi}{2} \end{cases}$$

$$\Rightarrow \begin{cases} \dfrac{3b}{2} & = & -\dfrac{3}{2} + 15i \\ a & = & \dfrac{i - 20 - bi}{2} \end{cases}$$

$$\Rightarrow \begin{cases} b & = & -1 + 10i \\ a & = & \dfrac{i - 20 - (-1 + 10i)i}{2} \end{cases}$$

$$\Rightarrow \begin{cases} b & = & -1 + 10i \\ a & = & \dfrac{i - 20 + i + 10}{2} \end{cases}$$

$$\Rightarrow \begin{cases} b & = & -1 + 10i \\ a & = & -5 + i \end{cases}$$

References